الجامعة السورية الخاصة
SYRIAN PRIVATE UNIVERSITY

Department of Informatics Engineering
System and Network Security Engineering

Graduation 1

كلية هندسة الذكاء الاصطناعي
FACULTY OF ARTIFICIAL INTELLIGENCE ENGINEERING

Cyber Attack Analysis Using Honeypot and Knowledge Graph

Prepared by :

Mayas AL- Refaie

Masa Daboul

Supervised by :

Dr. Wassim Juneidi
Eng.Ahmad Sheikha

Introduction

- What is the Honeypot ?
- Explain the concept of Knowledge Graph:
- What is the meaning of RAG ?

Unified Architecture: Integrating an SSH Honeypot, Large Language Models (LLMs), and Knowledge Graphs (KGs) into a single cohesive system.

Dynamic Interaction

Utilizing LLMs to generate realistic,context−aware responses, prolonging attacker engagement.

Retrieval−Augmented Generation (RAG):

Enhancing the LLM's capabilities by integrating RAG. This allows the honeypot to retrieve real-time threat intelligence (MITRE ATT&CK, CVEs) from external knowledge bases to generate highly accurate and informed responses.

Semantic Analysis (Knowledge Graph):

Transforming raw logs into a structured KG to map entities (Attackers, Tools, Techniques) and their hidden relationships.

Research Problem Map

Attack methodologies are increasingly distributed, stealthy and adaptive, which renders static defensive solutions insufficient.

PROBLEM	CONSEQUENCE	RESEARCH GAP
P1 Static, easily fingerprinted honeypot responses	Limited depth and quality of the collected attack data	**Research gap** No unified model integrates intelligent attacker interaction with structured knowledge representation.
P2 Event-level analysis in isolation from context	Attackers, tools, infrastructure and techniques remain uncorrelated; attack chains are not reconstructed	
P3 No systematic methodology from raw logs to machine-inferable knowledge	Logs stay unstructured, preventing automated reasoning and inference	

S	Pub Year	Authors	Title	Journal	Achievements	Features	Benefit in project
1	2026	Cotti et al.	OntoLogX: Ontology Guided Knowledge Graph Extraction From Cybersecurity Logs With Large Language Models	Advanced Intelligent Systems (Wiley)	Converts raw security logs into structured Knowledge Graphs using LLMs and links low-level evidence to MITRE ATT&CK tactics	Ontology-guided extraction; LLM-based pipeline; evidence-to tactic mapping	Defines the pipeline Honeypot → Logs → KG → Techniques; guides our ontology and ATT&CK mapping.
2	2026	Oliveira et al	An Integrated Honeypot and LLM Based Framework for Near Real-Time Detection and Behavioral Analysis of Malicious Activities	Journal of Cybersecurity and Privacy (MDPI)	Combined a Cowrie honeypot with a local LLM: Accuracy 94.23%, Precision 100%, F1 93.33%, detection within 25 s	Near real-time detection; deterministic preprocessing; session reconstruction	Justifies the stage Raw Logs → Preprocessing → Structured Events and Cowrie as our collection layer

S	Pub Year	Authors	Title	Journal	Achievements	Features	Benefit in project
3	2025	Li, Qiao, Le	Multi-View Intrusion Detection Framework Using Deep Learning and Knowledge Graphs	Information (MDPI)	Reached F1 = 0.963 on TON_IoT and 0.960 in a second setting.	Multi-view design; Knowledge Graph used as an additional view alongside deep learning.	Shows that relationship representation in a KG improves attack detection over isolated events.
4	2025	Authors not listed in source (to be verified)	Provenance-Based CTI Analysis with Knowledge Graph and RAG (KGRAG)	IEEE CSCloud (Cloud)	Identified threat actors with 95% accuracy and 100% provenance precision.	Knowledge Graph + RAG; provenance tracking of sources and evidence	Adds Source → Evidence → Attack; motivates evidence-linked triples and our RAG layer.

Figure 1: Proposed System Architecture

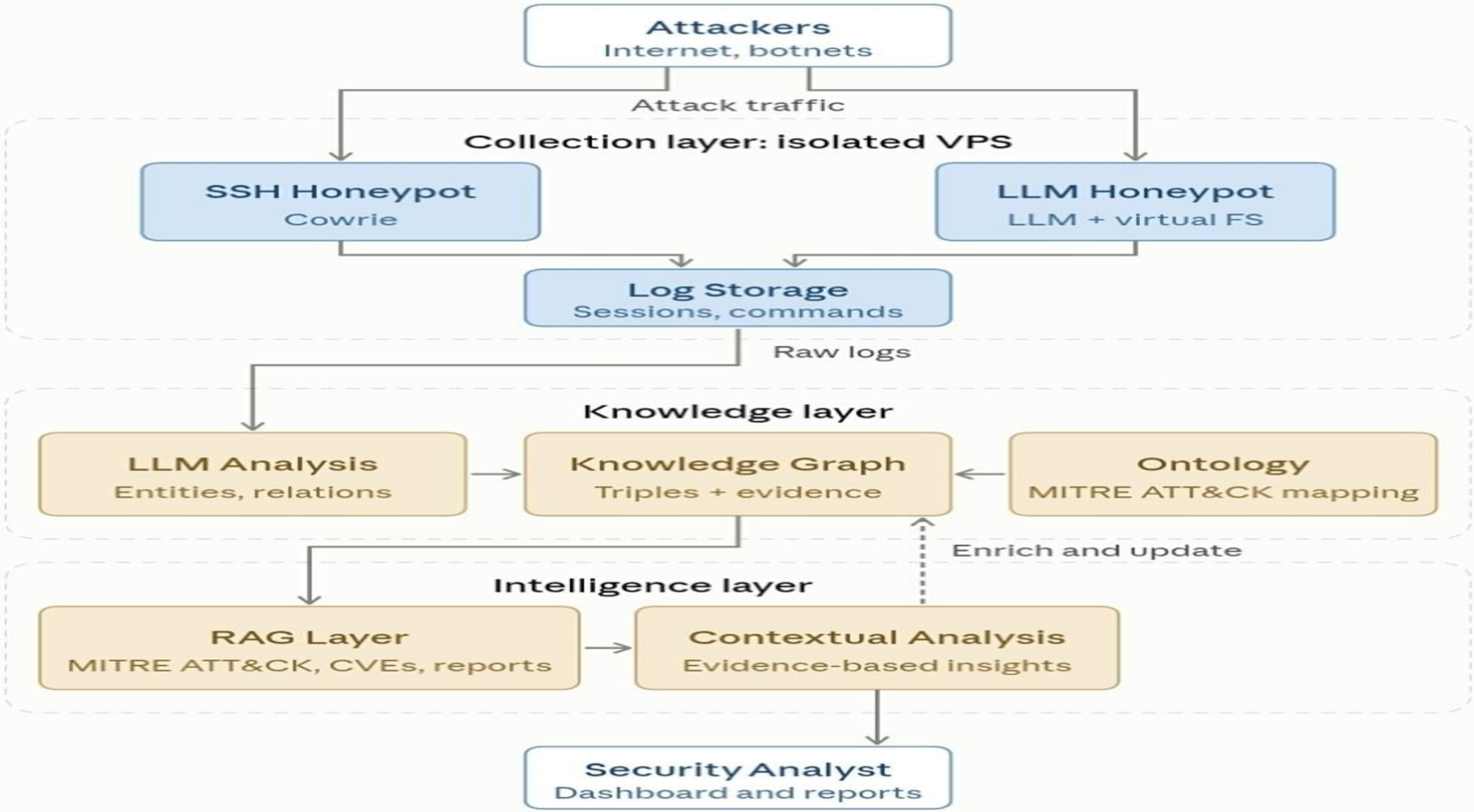


Figure 2: Collection Layer

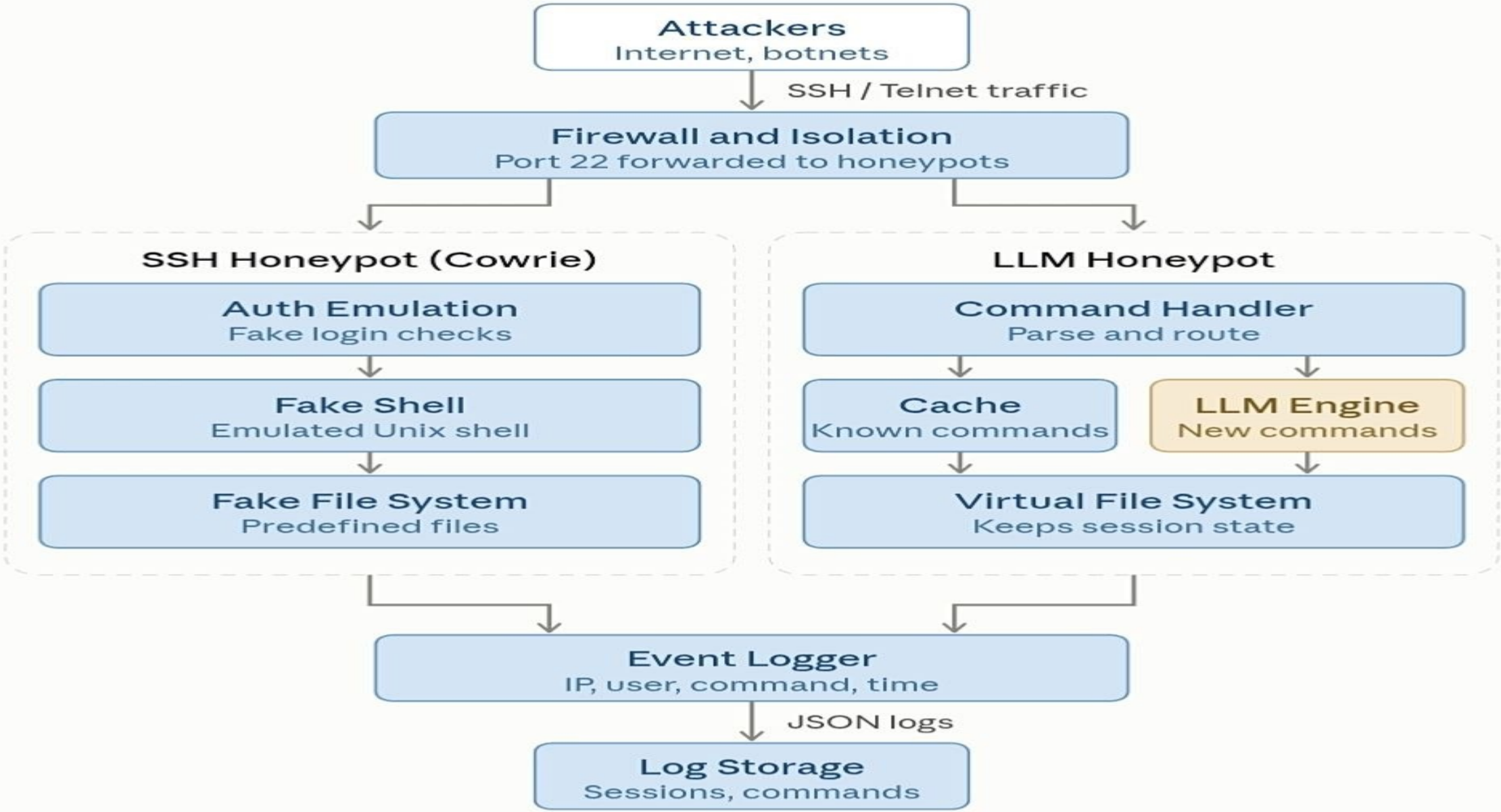


Figure 3: Knowledge Layer

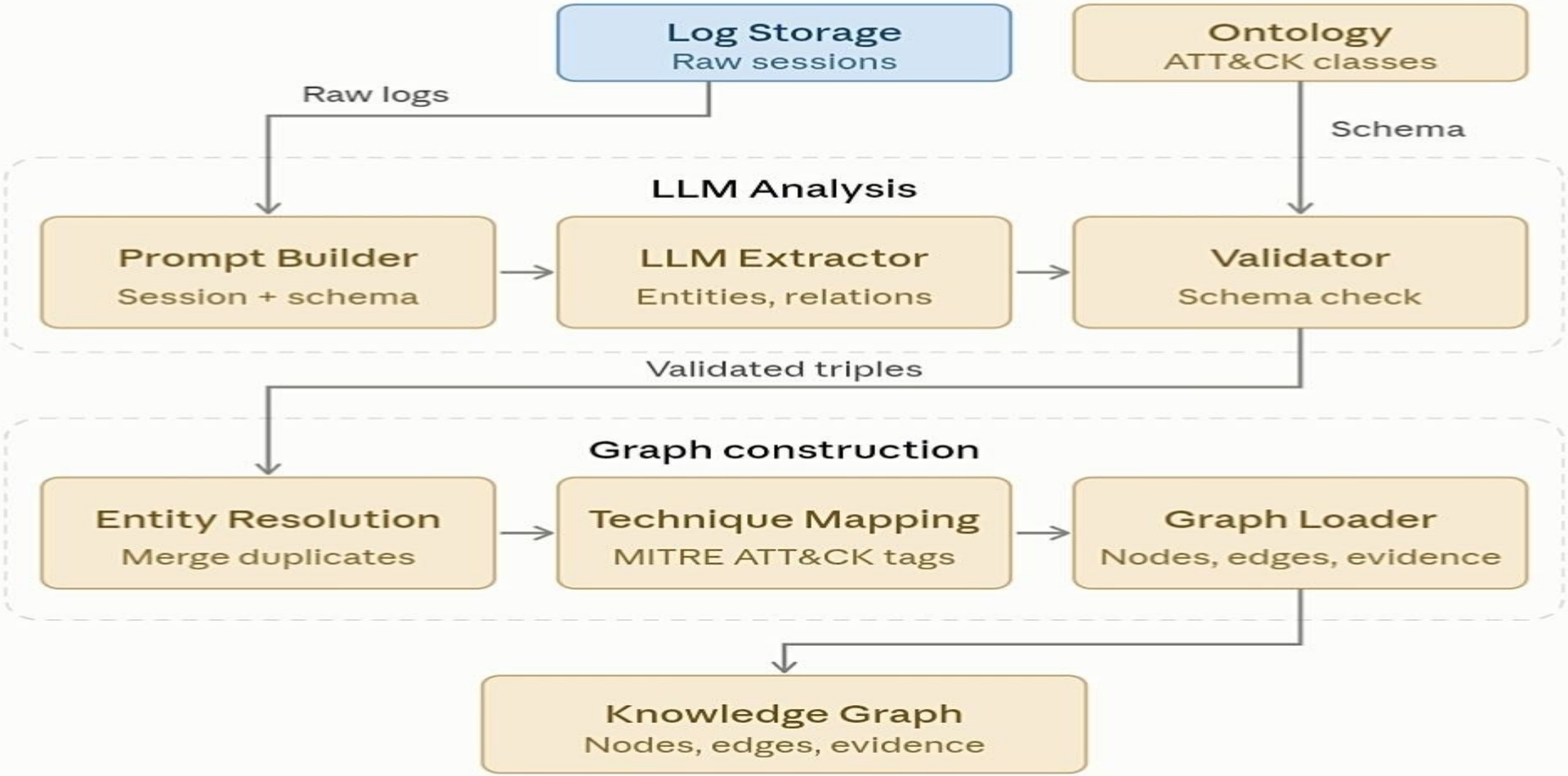


Figure 4: Intelligence Layer (RAG)

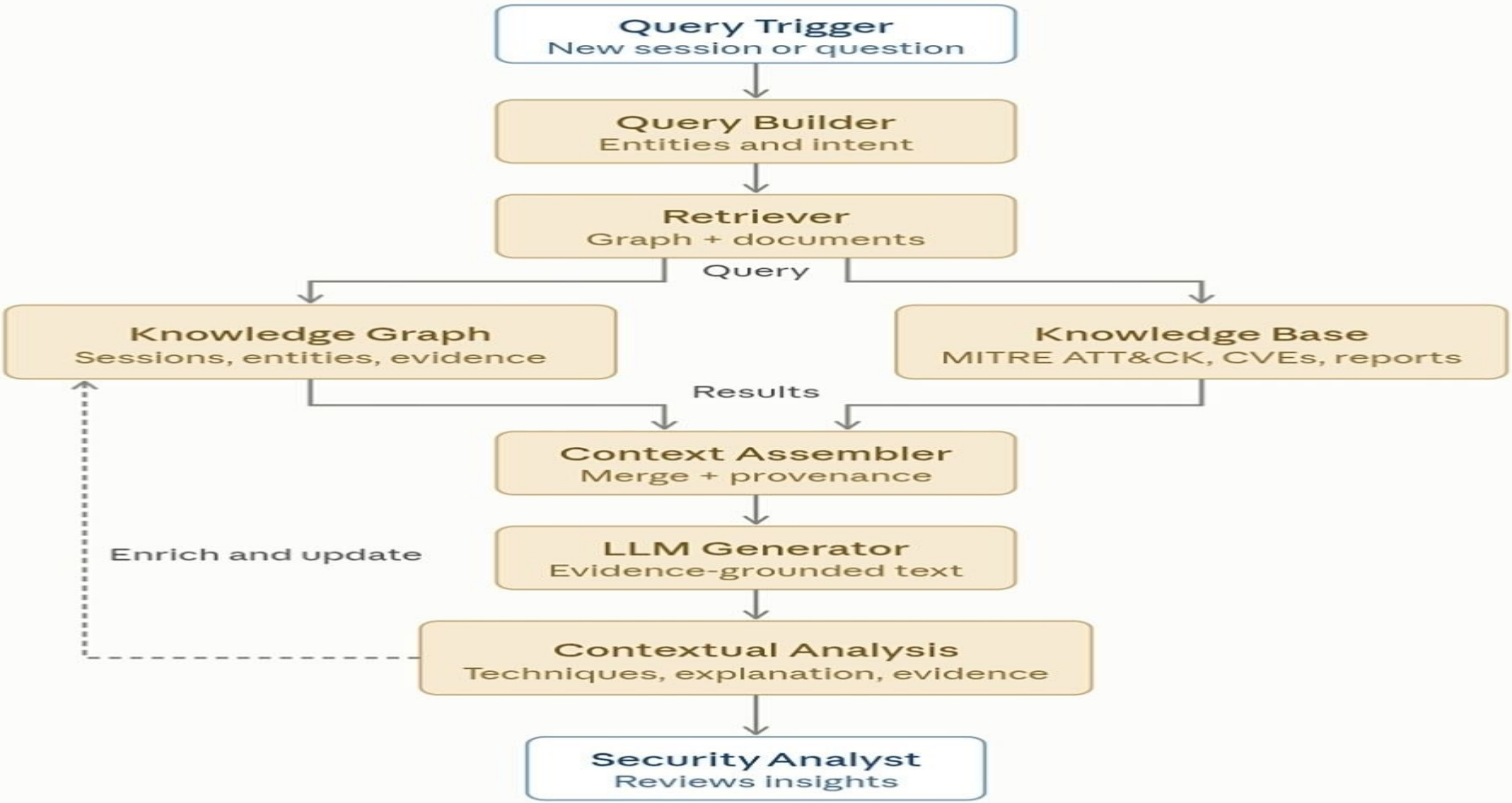
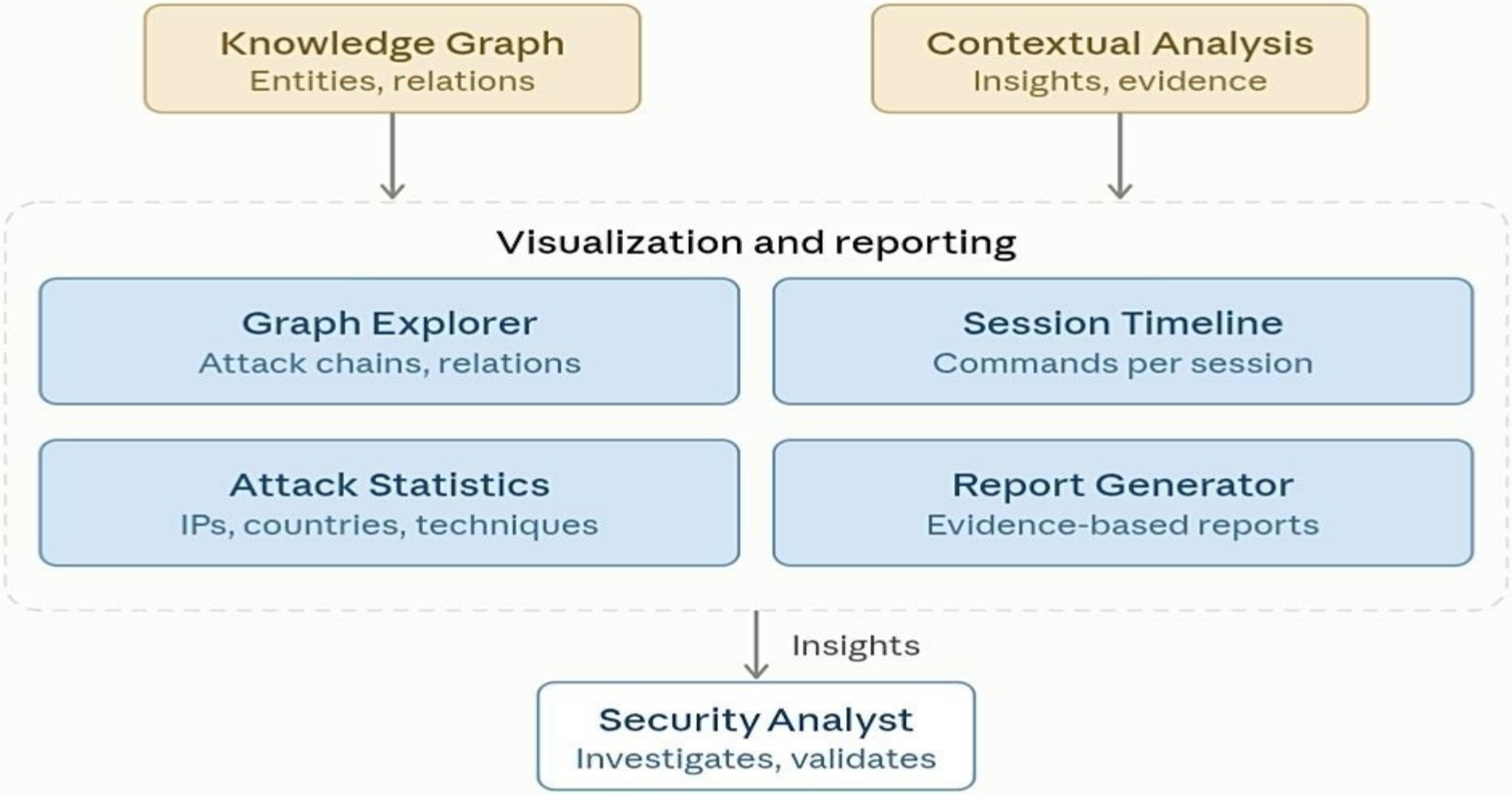
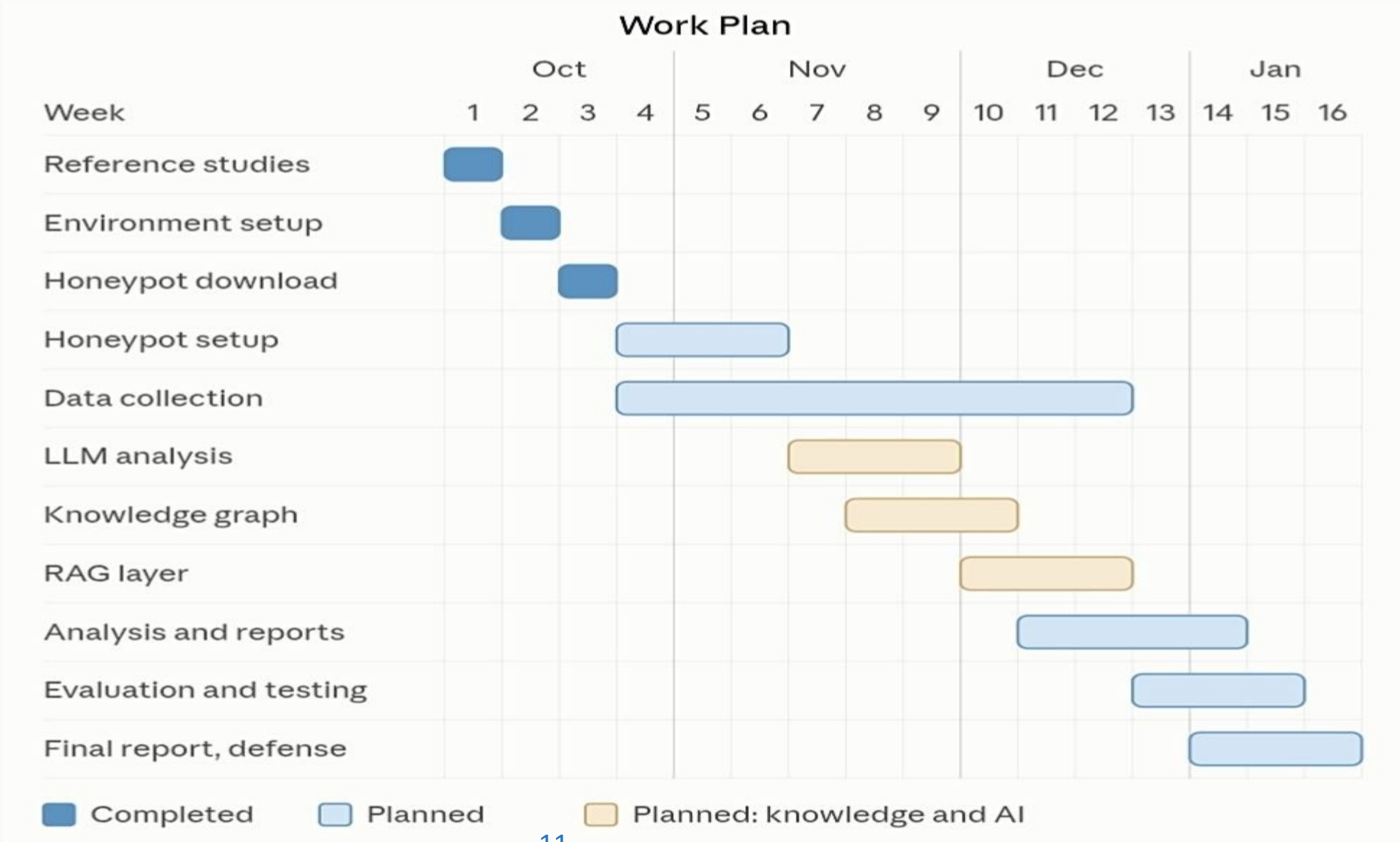


Figure 5: Visualization and Reporting




Work Plan
Oct
Nov
Dec
Jan
Week
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
Reference studies
Environment setup
Honeypot download
Honeypot setup
Data collection
LLM analysis
Knowledge graph
RAG layer
Analysis and reports
Evaluation and testing
Final report, defense
Completed
Planned
Planned: knowledge and AI

Work Environment & Tools

Virtualization & Operating System

- VMware Workstation
- Ubuntu 24.04 LTS

Purpose

To create an isolated virtual environment and run the honeypot safely.

Honeypots & Security Tools

- Cowrie (SSH Honeypot)
- HoneyAI Framework

Purpose

To simulate attacker interactions and collect security logs.

Programming & Data Processing

- Python
- Node.js
- JSON
- NetworkX

Purpose

To process logs, handle data, and build the knowledge graph.

AI & LLM Integration

- OpenAI API (ChatGPT)
- Ollama

Purpose

To enhance analysis capabilities using language models.

Threat Analysis Framework

- MITRE ATT&CK

Purpose

To map attacker behaviors to known techniques.

A secure, isolated, and intelligent environment for analyzing cyber attacks using honeypots and knowledge graphs.